

To: Tasha E. Fingerlin, PhD.
From: Opeyemi (Yemi) Jegede
Data: March 10, 2026
RE: Project 2 Preliminary Data Analysis Plan

Aim 1: The objective of this aim is to evaluate the temporal relationship between markers (measured from plasma samples) of peripheral inflammation and (a) memory consolidation, as well as (b) grey matter cortical thickness over two timepoints: baseline and one year later. Cognitive indices will be used to assess episodic memory function. Cortical thickness will be obtained from whole brain MRI scan after image processing. Table 1 below provides details about these markers and indices. We hypothesize that higher levels of inflammation markers will predict (a) decline in memory consolidation and (b) decrease in cortical thickness from baseline to one year. For each predictor and outcome variable, the change in values from baseline to one year after will be calculated and used for analyses such that each observation represents a unique subject. A multivariable linear regression model will be fitted to the data, and the regression coefficient and corresponding standard error will be used to perform each specified statistical hypothesis. Age at baseline and sex will be controlled for in this model by including them as covariates. Known covariates such as CVD risk, immunological health history conditions, and APOE genotype will be considered for inclusion in the model using a backwards model selection approach with a stringent significance level. Power calculation for each hypothesis test will be based on a simple linear regression model using a clinically meaningful effect elicited from the study principal investigator. This effect will be adjusted downward to account for the inclusion of covariates in the model since their inclusion may increase the variance of main effect.

Aim 2: The objective of this aim is to explore how inflammatory plasma marker levels and amyloid deposition interact together to predict clinical outcomes (cognitive indices and changes in brain structure) in aMCI. Amyloid deposition, a molecular proxy marker of AD pathology, will be measured using a Florbetapir-PET scan. Table 1 below provides details about the markers

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and variables. We hypothesize that there is an interaction between peripheral levels of inflammation and amyloid deposition, such that subjects with high levels of these marker pairs will have worse clinical outcome. Similar to Aim 1, a multivariable linear regression model will be fitted to the data for each statistical hypothesis, with the same consideration for inclusion of covariates described in Aim 1. Because this hypothesis involves an interaction term, the significance of each main effect will be tested and an interaction term included only if both main effects are significant at the chosen significance level. Power calculation will be based on the same approach as Aim 1. To power the interaction test, assumptions may be based on the binarized global SUVR because differences in effect are more easily specified for categorical than continuous variables. This approach (power calculation based on categorical variables) is more conservative (with respect to power) than continuous variables, so power will be higher when the continuous variable is used for analysis.

Table 1: Description of variables that will be measured in this study

Variable Type	Variable Category	*Specific Variables	Measurement Scale	Aims applicable to
Predictor	Peripheral inflammation markers	1) IL-6 (established) 2) TNF-alpha (established) 3) MCP-1 (novel) 4) Eotaxin-1 (novel)	Continuous	Aim 1 & 2
Predictor	Amyloid accumulation	Global SUVR (amyloid deposition)	Continuous	Aim 2
Outcome	Cognitive indices/measures of verbal & visual memory	1) 20' delay 2) Total correct recognition 3) D-Prime 4) 15' delay 5) Total correct 6) Proportion correct 7) 1-Week delay	Continuous	Aim 1 & 2
Outcome	Changes in brain structure	Grey matter cortical thickness	Continuous	Aim 1 & 2

**All the specified Specific Variables will be measured at both baseline and one year after*